Fixated but Failed to See: Recognition Errors in Visual Search

Jonathan Nir[1], Carmel R. Auerbach-Asch[1], Leon Y. Deouell[1,2]

[1] Edmond and Lily Safra Center for Brain Sciences, The Hebrew University of Jerusalem, Jerusalem 9190401, Israel
[2] Department of Psychology, The Hebrew University of Jerusalem, Jerusalem 9190501, Israel

Introduction:

- Visual search is a ubiquitous part of life, yet searchers sometimes fail to respond to clearly visible targets[1].
- Errors are commonly classified as **scanning, recognition** and **decision** errors, based on searcher's eye movements[2].
- Recognition errors, which occur despite searchers fixating the target, can indicate a failure in a variety of cognitive mechanism, including perception, attention, and memory[1, 3].
- Previously reported rates of fixated-but-missed errors vary remarkably across studies, from 2-34% depending on various stimulus properties[4-5].
- Here, we examine how **presentation mode**, **target category** and **target rotation** influence the probability of missing a fixated target ($P[\text{recognition error} \mid \text{on target}]$), while minimizing memory- and decision-related confounds.

Materials & Open Science:

Example trial with gaze data

Analysis code

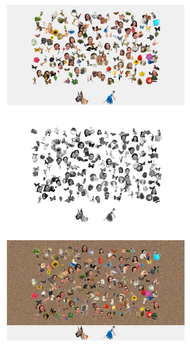
Data-sharing study (preprint)

References:

[1] Wolfe, J. M., Kosovicheva, A., & Wolfe, B. (2022). Normal blindness: When we look but fail to see. *Trends in Cognitive Sciences, 26*(9).
[2] Nodine, C. F., & Kundel, H. L. (1987). Using eye movements to study visual search and to improve tumor detection. *Radiographics, 7*(6).
[3] Nartker, M., Firestone, C., Egeth, H., & Phillips, I. (2023). Six ways of failing to see (and why the differences matter). *i-Perception, 14*(5).
[4] Godwin, H. J., Menneer, T., Riggs, C. A., Cave, K. R., & Donnelly, N. (2015). Perceptual failures in the selection and identification of low-prevalence targets in relative prevalence visual search. *Attention, Perception, & Psychophysics, 77*(1).
[5] Hout, M. C., Walenchok, S. C., Goldinger, S. D., & Wolfe, J. M. (2015). Failures of perception in the low-prevalence effect: Evidence from active and passive visual search. *Journal of Experimental Psychology: Human Perception and Performance, 41*(4).

Methods:

Stimulus



- 180 icons in array
- All trials have 1-3 targets
- Each specific target appears 0-2 times
- 3 trial types
- 6 target categories
  Human Face, Human Body Part, Animal Face, Animal body, Hand-Made Object, Natural Object
- Target rotation $[-20°, 20°]$

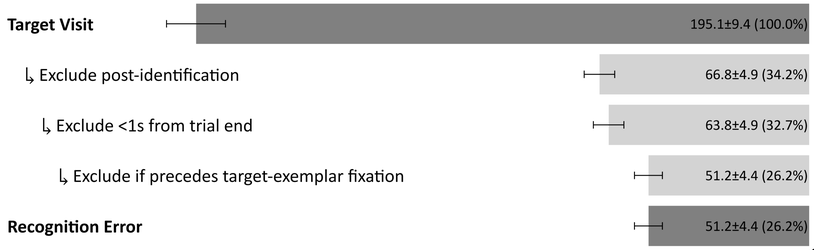
Participants & Apparatus

- $N = 12$ Participants
- 60 trials/participant (10/block)
- Tobii Pro Spectrum @ $600Hz$

Hierarchical Bayesian Model

$$P[\text{recognition error}] \sim C(\text{trial type}) \cdot C(\text{target category}) \cdot |\text{target angle}| + (1 \mid \text{subject})$$

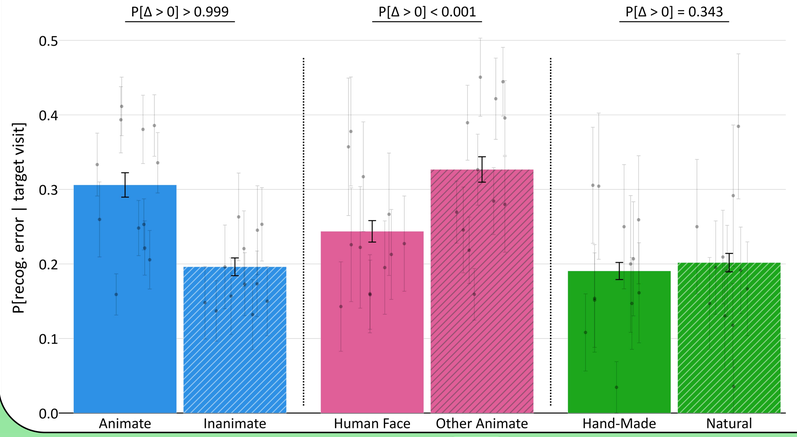
Defining Recognition Errors



Results:

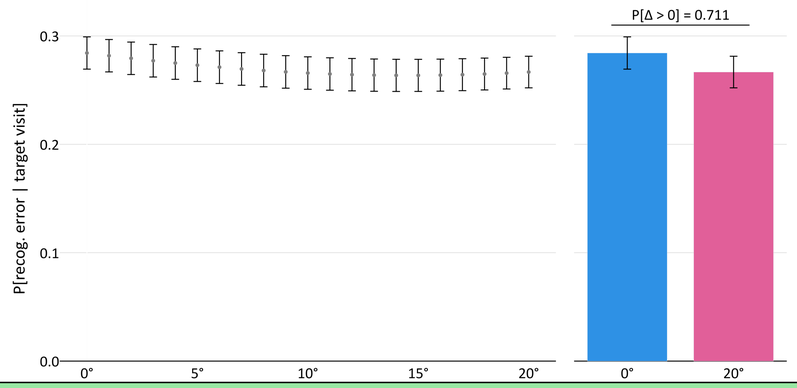
Semantic Effect

- **Animate** targets were less recognized than **inanimate** targets.
- Among **animate** targets, **human faces** were more recognized.
- No difference between **hand-made** and **natural** objects.



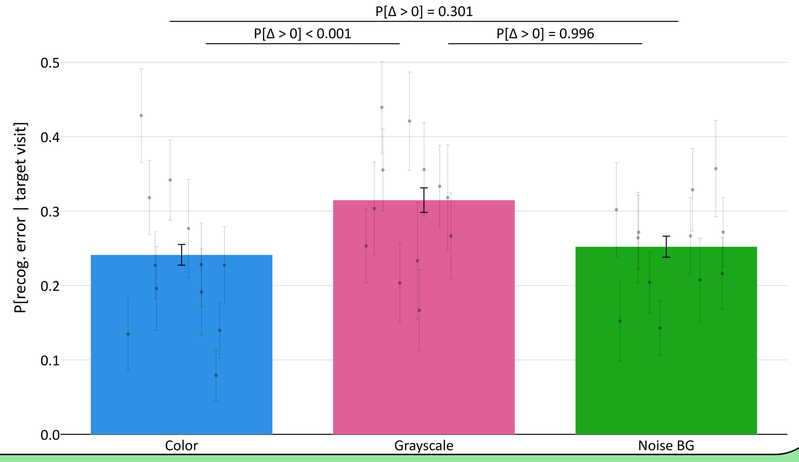
Rotation Effect

- Target angles collapsed over counter- and clock-wise rotation
- No difference between **upright** (0°) and **rotated** (±20°)



Visual Format Effect

- Grayscale images yielded lower recognition than color images.
- Background noise did not impair recognition.



jonathan.nir@mail.huji.ac.il @JonNir3 @NeuroJon